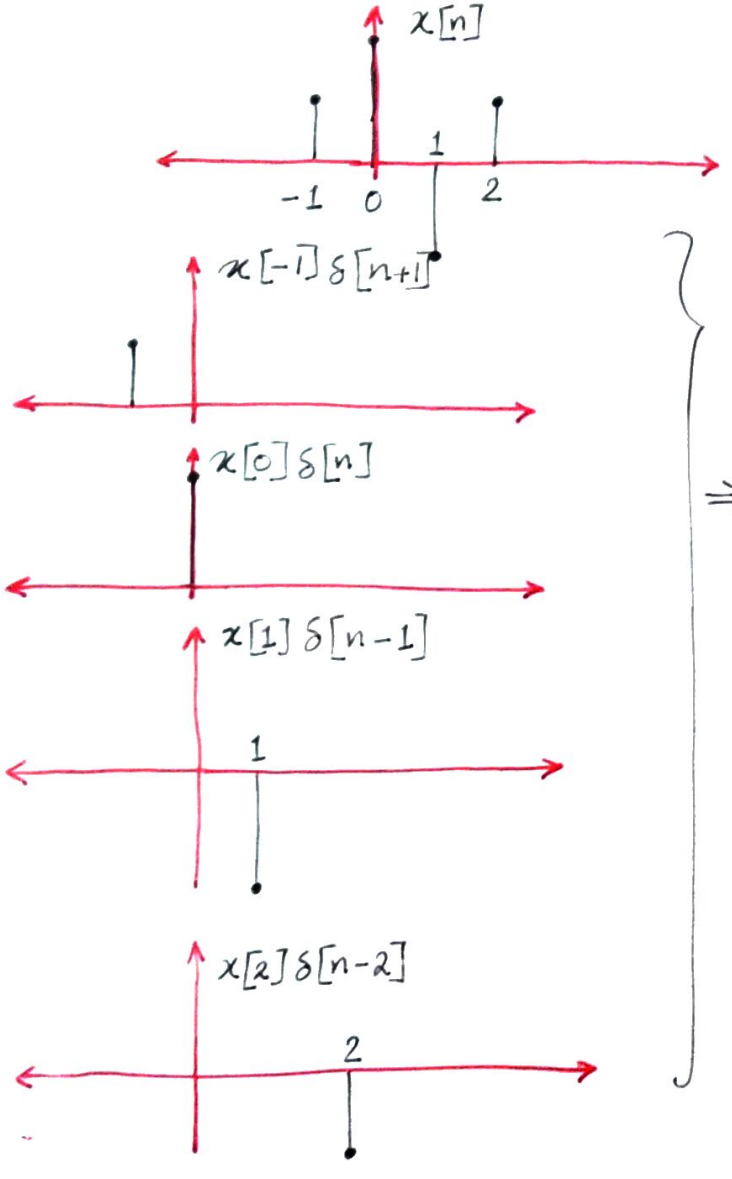
Linear Time-Invariant Systems

Discrete-time LTI Systems :

(1) Representation of discrete-time signals in terms of impulses



$$\Rightarrow \quad x[n] = x[-1]\,\delta[n+1] + x[0]\,\delta[n] + x[1]\,\delta[n-1] + x[2]\,\delta[n-2] + \cdots$$

$$= \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

Shifting property of discrete-time unit impulse

(2) Discrete-time unit impulse response and the convolution-sum representation of LTI systems :

Let $\mathcal{H}$ be the operator which denote the system to which $x[n]$ is applied. Then the output is given by -

$$y[n] = \mathcal{H}\left\{\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]\right\}$$

$$= \sum_{k=-\infty}^{\infty} \mathcal{H}\left\{x[k]\,\delta[n-k]\right\} \quad \text{(linearity-superposition)}$$

$$\boxed{y[n] = \sum_{k=-\infty}^{\infty} x[k]\,\mathcal{H}\left\{\delta[n-k]\right\}}$$

($\because$ n is the time-index, the quantity $x[k]$ is a constant w.r.t. system operator)

The system output is weighted sum of the response of the system to time shifted impulses — for linear system.

For time-invariant systems if $\mathcal{H}\{\delta[n]\} =: h[n]$. Then

$$\boxed{y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}$$

The system output is weighted sum of the time-shifted impulse responses.

$$\boxed{x[n] * h[n] := \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}$$

An LTI system is completely characterized by its response to a single signal, namely to the unit impulse.

$$x[n] = \alpha^n u[n] \qquad y[n] = \sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha} \quad \text{for } n \geq 0$$

$$h[n] = u[n]$$

e.g. Consider the discrete-time LTI system model representing a two-path propagation channel have an output

$$y[n] = x[n] + \frac{1}{2} x[n-1]$$

What is the impulse response?

<u>Sol^n:</u> with $x[n] = \delta[n]$

$$y[n] = \delta[n] + \frac{1}{2}\delta[n-1]$$

$$\Rightarrow y[n] = \begin{cases} 1, & n = 0 \\ 1/2, & n = 1 \\ 0, & \text{otherwise.} \end{cases}$$

Determine the output of system corresponding to input

$$x[n] = \begin{cases} 2, & n = 0 \\ 4, & n = 1 \\ -2, & n = 2 \\ 0, & \text{otherwise} \end{cases}$$

Here $x[n] = 2\delta[n] + 4\delta[n-1] - 2\delta[n-2]$

$$\therefore y[n] = 2\delta[n] + \delta[n-1] + 4\delta[n-1] + 2\delta[n-2] - 2\delta[n-2] - \delta[n-3]$$

$$= -\delta[n-3] + 5\delta[n-1] + 2\delta[n]$$

For short impulse responses and signals it is easier to find the output corresponding to the input. However for signals which are longer, this procedure won't be easy.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Note that once n is fixed, we will have the o/p by shifting the impulse response k units in time and multiplying by $x[k]$.

So, the procedure is —

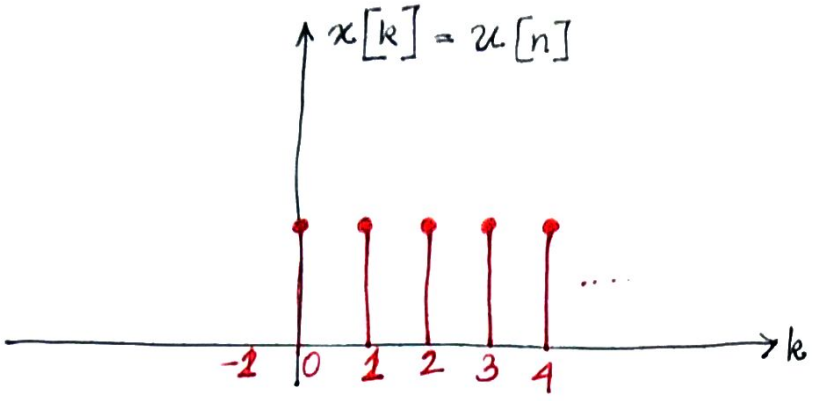
(1) Plot both $x[k]$ and $h[n-k]$ as a function of k.
To determine $h[n-k]$, first reflect $h[k]$ about $k=0$ — we get $h[-k]$. The shift by $-n$.

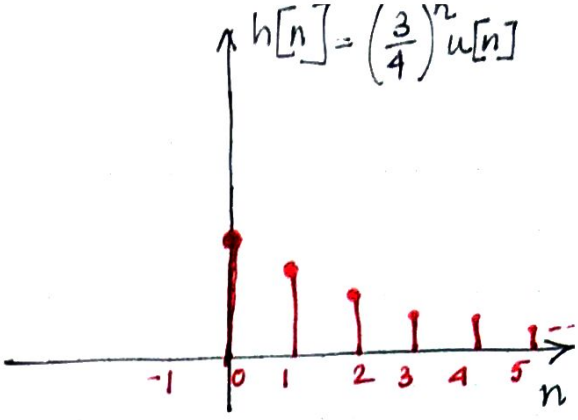
(2) Begin with large negative n. — Shift $h[-k]$ far left.

(3) Increase the shift n until the mathematical representation for $w_n[k]$ changes. The value of n at which the change occurs defines the end of the current interval and beginning of a new one

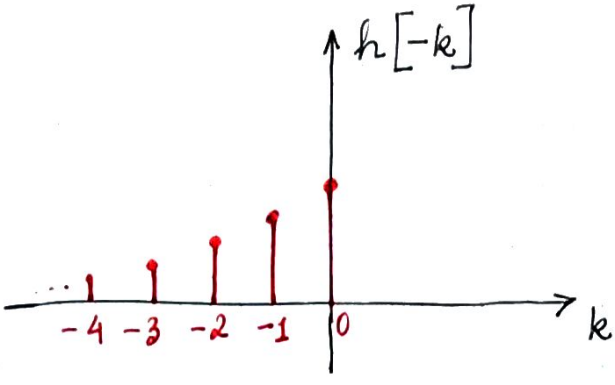
(4) Let n be the new interval — Repeat steps 3 & 4 until all intervals of time shifts and corresponding mathematical formula for $w_n[k]$ is figured out.

(5) For each interval of time shift, sum all $w_n[k]$ to get $y[n]$

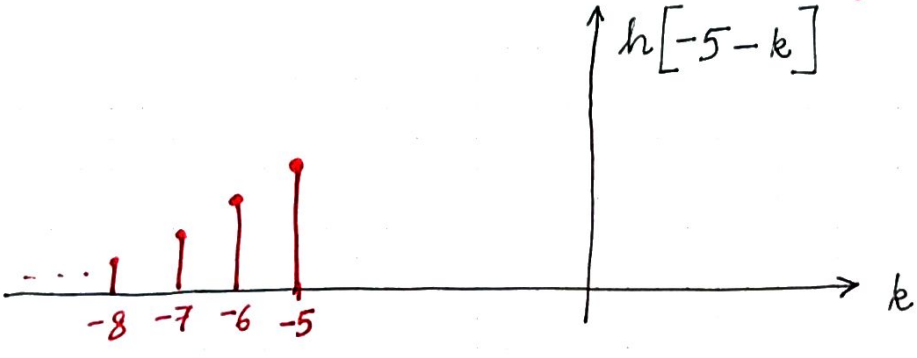


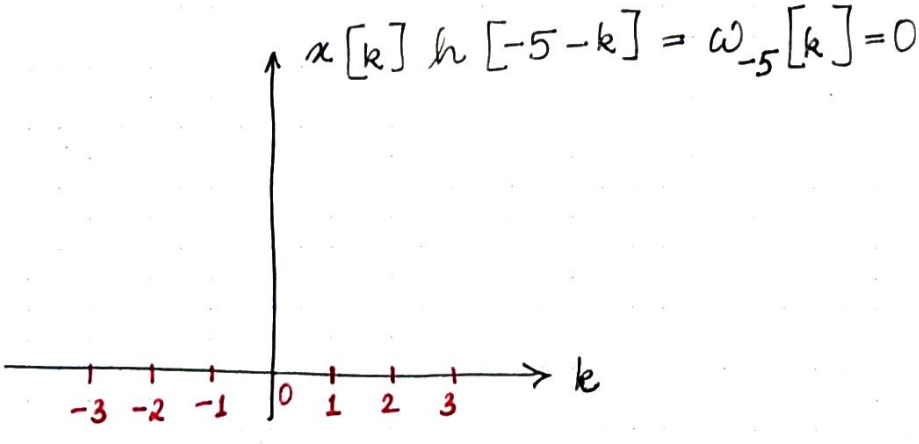


– Reflect

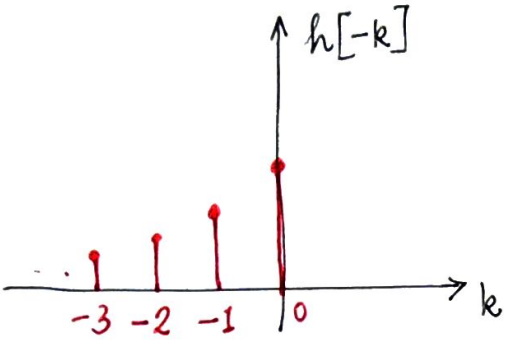


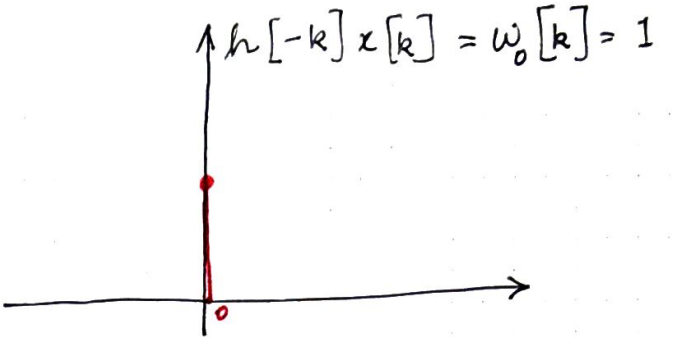
– Shift to the left with large negative n. (Say $n=-5$)



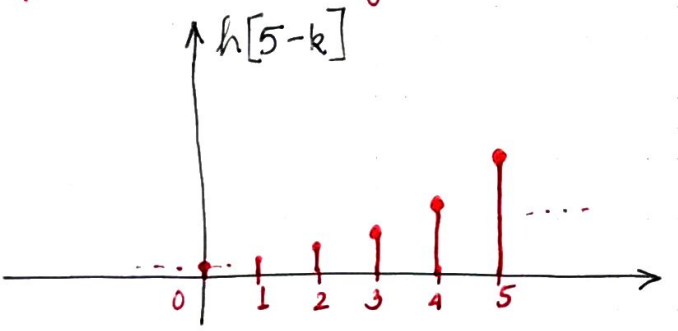


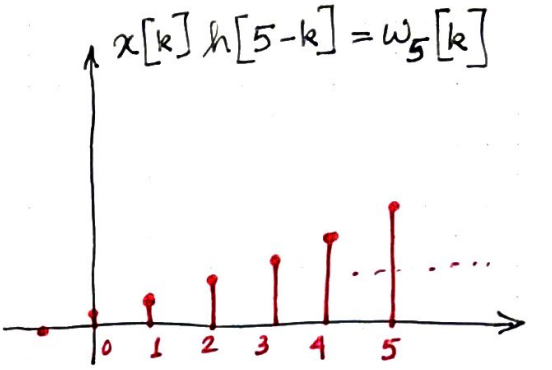
- Shift to the right (say $n=0$)





- Shift more to right (say $n=5$)





$$\Downarrow$$

$$w_5[k] = \left(\frac{3}{4}\right)^{5-k}, \quad 0 \leq k \leq 5$$

- Shift more to right (say $n=10$)

$$w_{10}[k] = \left(\frac{3}{4}\right)^{10-k}, \quad 0 \leq k \leq 10$$

$$\therefore \quad w_n[k] = \begin{cases} \left(\frac{3}{4}\right)^{n-k}, & 0 \leq k \leq n \\ 0, & k < 0. \end{cases}$$

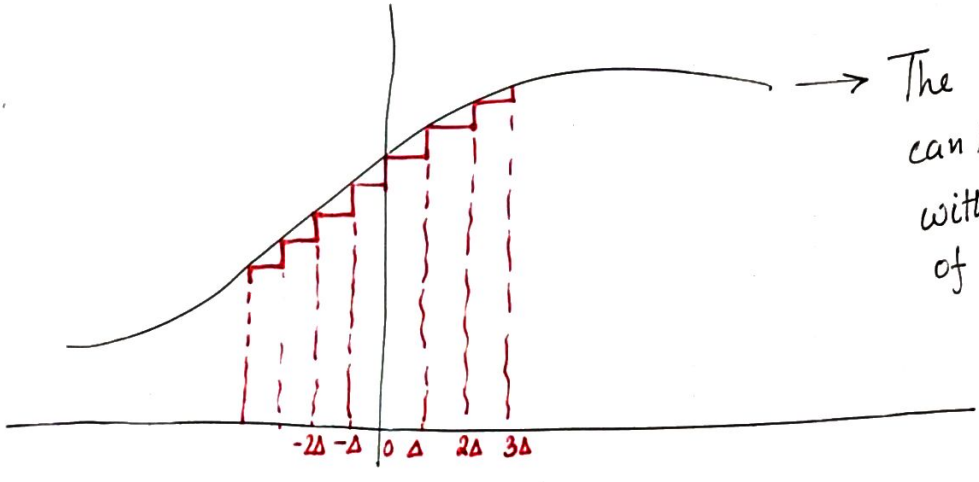
$$\therefore \quad \boxed{y[n] = \sum_{k=-\infty}^{\infty} w_n[k]}$$

The convolution integral:

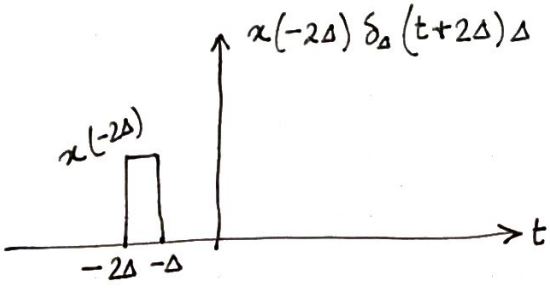
→ First express the signal as the weighted superposition of time-shifted impulses:

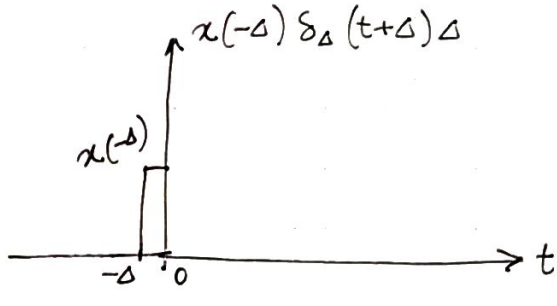
$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau}$$

First we consider a stair-case approximation of the signal



→ The signal can be approximated with a combination of delayed pulses.





The approximate signal can be written as a linear combination of delayed pulses. If we define

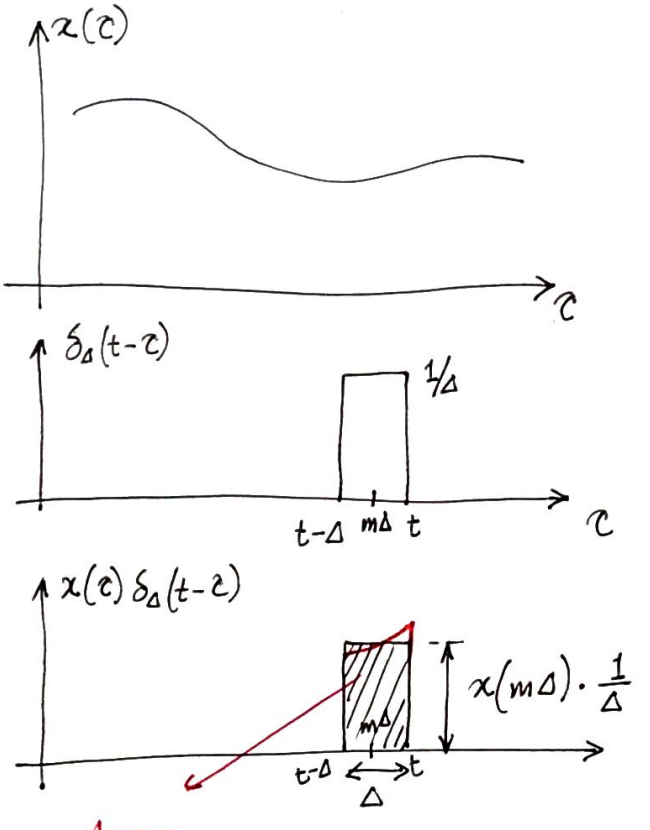
$$\delta_\Delta(t) = \begin{cases} \dfrac{1}{\Delta}, & 0 \leq t < \Delta \\ 0, & \text{otherwise} \end{cases}$$

$\because$ $\delta_\Delta(t)\Delta$ has unit amplitude, we have

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta)\,\delta_\Delta(t-k\Delta)\Delta$$

As we let Δ approach 0, the approximate $\hat{x}(t)$ becomes better and better and in the limit we have

$$x(t) = \lim_{\Delta \to 0} \sum_{k=-\infty}^{\infty} x(k\Delta)\,\delta_\Delta(t-k\Delta)\Delta$$



Area of the shaded region approaches the area under $x(\tau)\,\delta_\Delta(t-\tau)$ as $\Delta \to 0$.

Area =
$x(m\Delta)\cdot\frac{1}{\Delta}\cdot\Delta = x(m\Delta)$

$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau}$$

Shifting integral.

$$y(t) = \mathcal{H}\{x(t)\} = \mathcal{H}\left\{\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau\right\}$$

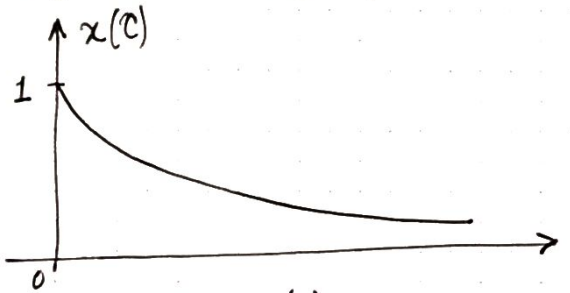
$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)\,\mathcal{H}\{\delta(t-\tau)\}\,d\tau}$$

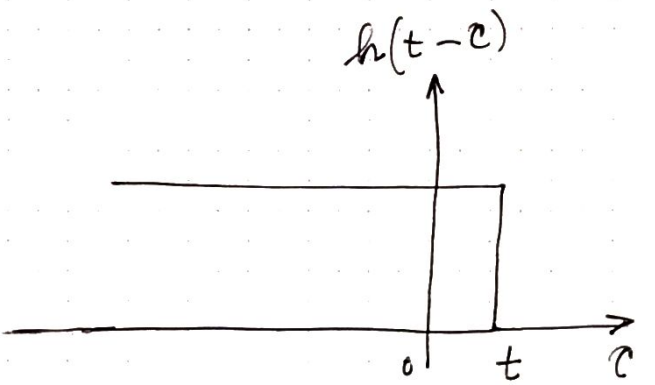
For linear system

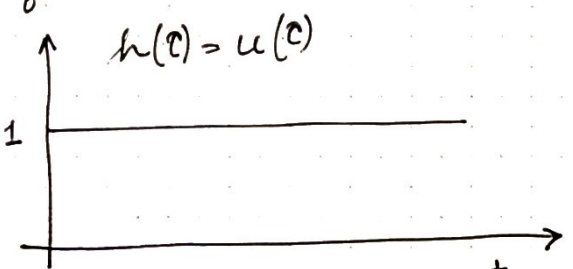
For time invariant case: $\mathcal{H}\{\delta(t-\tau)\} = h(t-\tau)$

$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau}$$

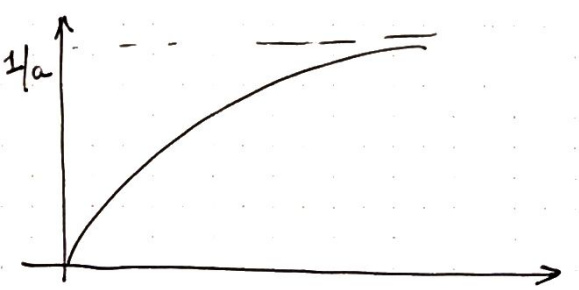
e.g. $x(t) = e^{-at}u(t)$, $a > 0$







$$y(t) = \int_0^t e^{-a\tau}\,d\tau = \left.\frac{e^{-a\tau}}{-a}\right|_0^t = \frac{1}{a}\left(1 - e^{-at}\right)$$



Properties of Linear Time-Invariant Systems:

$$\boxed{\begin{aligned} y[n] &= \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k] = x[n] * h[n] \\ y(t) &= \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau = x(t) * h(t) \end{aligned}}$$

→ The unit impulse response does not completely characterize the behavior of a non-linear system.

e.g.
$$h[n] = \begin{cases} 1, & n = 0, 1 \\ 0, & \text{otherwise} \end{cases} \longrightarrow \text{(A)}$$

If the system is LTI then (A) characterizes all the i/o behavior.

e.g. $y[n] = x[n] + x[n-1]$

On the other hand, there are many nonlinear systems with the same response. e.g.

$$y[n] = \left(x[n] + x[n-1]\right)^2$$

$$y[n] = \max\left(x[n], x[n-1]\right).$$

— LTI systems have many properties that other systems do not posses.

The Commutative Property

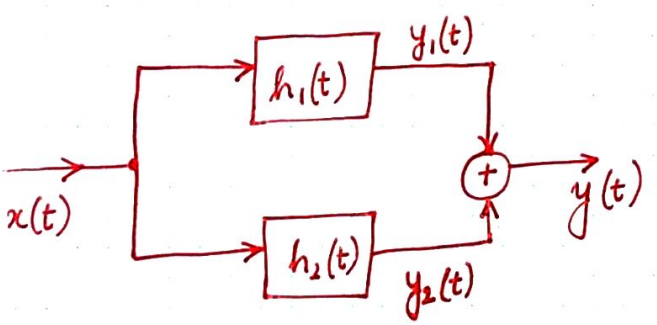
The output of an LTI system with input $x[n]$ and unit impulse response $h[n]$ is identical to the output of the LTI system with input $h[n]$ and unit impulse response $x[n]$.

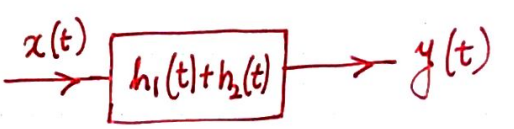
$$\boxed{\begin{aligned} x[n] * h[n] &= h[n] * x[n] \\ x(t) * h(t) &= h(t) * x(t) \end{aligned}}$$

The Distributive property

$$x[n] * \left(h_1[n] + h_2[n]\right) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$x(t) * \left[h_1(t) + h_2(t)\right] = x(t) * h_1(t) + x(t) * h_2(t)$$





e.g. $x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-n]$

$h[n] = u[n]$